

EVALUATING EXPRESSIONS WITH RADICALS



LESSON INTRODUCTION

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers, including exponents and radicals.

LEARNING TARGETS

- I can solve problems involving the order of operations with rational numbers, including exponents and radicals.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers, including grouping symbols, whole-number exponents, and absolute value.

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- On which step should radicals be evaluated when performing the order of operations?
- How do I evaluate an expression having radicals using the order of operations?

LESSON NARRATIVE

In this lesson, students build on their knowledge of using the order of operations in solving problems to include radicals. Students explore expressions with radicals to learn that radicals and exponents are evaluated during the same step when performing the order of operations. Students learn a radical cannot be evaluated until the value under the radical has been determined. Students engage in analyzing student work and repeated practice to build mathematical fluency in solving problems involving the order of operations with radicals.

COMPONENT SUMMARY

12.7.1 WARM-UP (~5 MIN.)

Create a square root expression

12.7.3 GUIDED INSTRUCTION (15-25 MIN.)

Evaluate expressions with radicals

12.7.5 YOUR TURN! (10 MIN.)

Find the value of expressions with radicals

12.7.2 REFRESHER (5 MIN.)

Apply the order of operations

12.7.4 GUIDED PRACTICE (10-15 MIN.)

Find the value of expressions with radicals

12.7.6 WRAP-UP (~5 MIN.)

Reflect on level of interest and understanding in the learning

target

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Radical
- Absolute Value
- Grouping Symbols
- Square Root
- Cube Root

MATERIALS

- (Optional) Desmos scientific calculators

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may think finding the cube root of a number means dividing by 3.
- When evaluating fractions with expression in the numerator and/or in the denominator, students may not understand that the entire expression in the numerator must be evaluated first.
- Students may not think the order of operations matters when entering an expression into the calculator.
- Students may not remember absolute value symbols are grouping symbols.

THINGS TO CONSIDER

- If time is an issue, then consider assigning 12.7.5 Your Turn! for homework.
- While students will not have a calculator on the Florida Statewide Assessment (FSA) for this benchmark, it is helpful for them to see that you have to group the operations in the numerator. For an example, reference 12.7.4 Guided Practice question 1.

LITERACY AND LANGUAGE ROUTINES

Critique, Correct, and Clarify

In 12.7.3 Guided Instruction, students engage in the Critique, Correct, and Clarify language routine. In question 3, students are asked to critique the mathematical work of three students. As part of their critique, students identify the differences in the steps used to evaluate the expression. Students then identify whose work was correct and clarify the errors made by the other two students to explain why they resulted in an incorrect answer.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Mathematical fluency requires being flexible and precise while carrying out calculations. By providing students with multiple opportunities to execute procedures and perform calculations, this lesson allows students to maintain flexibility and build towards mathematical fluency in accurately performing the order of operations on expressions with rational numbers, radicals, and exponents.

DIFFERENTIATE INSTRUCTION

Question 3 in 12.7.3 Guided Instruction can provide an opportunity for both auditory and visual entry points. Consider providing roles for students for the questions. For example, one student might illustrate and circle the errors in the work, and one might verbally explain the errors. If students are stuck on question 3d, consider prompting them to work the problem on their own to compare their work with the students who made errors. While one of the students did correctly evaluate the expression, it can be helpful for students to engage in the work of evaluating the expression themselves to better connect with the context and understand where of the two students made mistakes.

12.7.1 WARM-UP: SQUARE ROOT EXPRESSION

TEACHER MOVES

Students learned to evaluate radicals in Unit 2 and used this skill again in Unit 5 with Pythagorean Theorem. They will be evaluating radicals within the lesson, so 12.7.1 Warm-Up provides an opportunity to revisit the concept if observations of student work warrant the need.



12.7.1 Warm-Up: Square Root Expression

- Place a digit in each blue box to create an expression with a value as close to 0 as possible. Use only the digits 0 through 9, at most one time each.

Answers will vary.

$$\sqrt{\begin{array}{|c|c|} \hline 2 & 5 \\ \hline \end{array}} - \sqrt{\begin{array}{|c|c|} \hline 3 & 6 \\ \hline \end{array}} + \sqrt{\begin{array}{|c|} \hline 1 \\ \hline \end{array}}$$

- Describe your strategy in determining where to put the numbers.

Answers will vary. I used the smallest square roots possible to reach -1, and then added the square root of 1.

- What is the value of your expression?

0

- Create another expression that is as close or closer to 0 than your first.

Answers will vary. $\sqrt{36} - \sqrt{49} + \sqrt{1}$

**12.7.2 Refresher: Order of Operations**

Recall that the order of operations is an agreed-upon set of rules for how to evaluate an expression.

1. Work with your partner to number each operation so that the operations are in the correct order.

- | | |
|---|--|
| 3 | Perform multiplication and division in order from left to right. ($\rightarrow$) |
| 2 | Evaluate expressions with exponents. |
| 1 | Perform operations inside grouping symbols. |
| 4 | Perform addition and subtraction in order from left to right. ($\rightarrow$) |

2. Based on what you have learned about radicals, discuss with your partner where you think evaluating radicals falls within the order of operations. Then, write a summary of your discussion.

Answers will vary. We think radicals will fall right after evaluating expressions with exponents because radicals are the inverse of exponents.

12.7.2 REFRESHER: ORDER OF OPERATIONS**TEACHER MOVES**

Grade 8 is the first time evaluating radicals is included within the order of operations because they first learn to evaluate square roots and cube roots in this grade. Many students will recognize that because there is an inverse relationship between radicals and exponents, these will be in the same step (similar to multiplication/division and addition/subtraction). However, there needs to be an opportunity for them to discuss this and then formalize the idea.

Teachers should listen to student discussions and use their thinking to build discussion toward recognizing that radicals are evaluated in the same step as evaluating exponents. This provides an opportunity for students to consider this, and many will come to this conclusion before being told. Consider posting an example to assist with the discussion.

12.7.3 GUIDED INSTRUCTION: EVALUATING EXPRESSIONS WITH RADICALS

TEACHER MOVES

Allow students the opportunity to review the steps Simon took and have a thorough discussion with their partner. Monitor the discussions to listen for what students say about the different grouping symbols. If students do not remember absolute value symbols and what they represent, have a classmate briefly share why -2 is +2 when the absolute value symbols are removed.

TEACHER MOVES

Depending on students' foundational understanding, they may not remember absolute value symbols are a type of grouping symbol. Be prepared to reinforce this if you find students do not identify it in their answer to question 1b.



12.7.3 Guided Instruction: Evaluating Expressions With Radicals

Evaluating exponents and radicals falls within the same step in the order of operations, where each is evaluated as from left to right, after operations within grouping symbols are performed.

- Simon evaluated the expression $\frac{(3-\sqrt{25})^3}{16}$, as shown.

Original Expression	$\frac{(3-\sqrt{25})^3}{16}$
Step 1	$\frac{ 3-5 ^3}{16}$
Step 2	$\frac{ -2 ^3}{16}$
Step 3	$\frac{2^3}{16}$
Step 4	$\frac{8}{16} = \frac{1}{2}$

- With your partner, discuss the steps Simon took to find the value of the expression.
- Which grouping symbols are used in the expression?
parentheses and absolute value
- Use the words or phrases in the bank to complete the statements.

absolute value	equivalent	exponent
grouping symbols	radicals	subtraction

Simon started by evaluating $3 - \sqrt{25}$ because the absolute value bars are *grouping symbols*. Within the bars, Simon first evaluated $\sqrt{25}$ because evaluating *radicals* comes before *subtraction*. He was able to evaluate the *exponent* only after finding the value of the grouped *absolute value* expression. Finally, he found an *equivalent* fraction for the expression that remained.

2. Find the value of the expression. Show your work.

$$\begin{aligned} & \left(\frac{1}{3}\right)^{-2} + \sqrt{4^2 + 9} \\ &= \left(\frac{3}{1}\right)^2 + \sqrt{4^2 + 9} \\ &= 3^2 + \sqrt{16 + 9} \\ &= 9 + \sqrt{25} \\ &= 9 + 5 \\ &= 14 \end{aligned}$$

3. Simon, Tonya, and Cassandra evaluated the following expression using substitution and the order of operations. Their work is shown in the table.

$$\sqrt{56.5 - b} - 4.1^2, \text{ where } b = -7.5$$

Simon	Tonya	Cassandra
$\sqrt{56.5 - b} - 4.1^2$	$\sqrt{56.5 - b} - 4.1^2$	$\sqrt{56.5 - b} - 4.1^2$
$\sqrt{56.5 - 7.5} - 4.1^2$	$\sqrt{56.5 + 7.5} - 4.1^2$	$\sqrt{56.5 + 7.5} - 4.1^2$
$\sqrt{49} - 4.1^2$	$\sqrt{64} - 4.1^2$	$\sqrt{64} - 4.1^2$
$7 - 4.1^2$	$8 - 4.1^2$	$8 - 4.1^2$
$7 - 16.81$	$8 - 16.81$	$8 - 16.81$
-9.81	24.81	-8.81

- Discuss each student's work with your partner.
- Circle the steps where you notice differences in the students' work.
- Which student correctly evaluated the expression?
Cassandra

12.7.3 GUIDED INSTRUCTION: EVALUATING EXPRESSIONS WITH RADICALS (CONTINUED)

TEACHER MOVES

Students could be given the opportunity to discuss the expression and where they begin through a turn and talk or other strategy. However, rather than releasing this to the students, the intention is to model how to evaluate the expression using the order of operations through a think-aloud.

Explain that you notice both exponents and a radical in this expression when reading it from left to right. The exponent $\left(\frac{1}{3}\right)^{-2}$ would be evaluated before the radical, but the radical can't be evaluated until the value under the radical is determined. In this sense, the radical acts as a grouping symbol. Under the radical, 4^2 is evaluated before adding 9 to it because exponents are evaluated before addition. The resulting expression is $\left(\frac{1}{3}\right)^{-2} + \sqrt{25}$. Here, the exponent and radical would be evaluated and then the two would be added together, resulting in a value of 14.

TEACHER MOVES

Evaluating using the order of operations after substitution is included explicitly in the benchmarks in 6th grade. It has been included in Math Nation resources in both 7th and 8th grade when addressing order of operations for two reasons:

- Both the 7th and 8th grade benchmarks refer to solving mathematical problems, which evaluating an algebraic expression is included in that.
- Continuing this through middle grades provides an opportunity to solidify an understanding of variables as representing numbers AND sets students up to be able to evaluate a variety of function types when they reach Algebra 1 that would not be there if this concept was not continued throughout all three grades.

12.7.3 GUIDED INSTRUCTION: EVALUATING EXPRESSIONS WITH RADICALS (CONTINUED)

ENRICHMENT

- How did Simon making an error so early in evaluating the expression affect his work moving forward?
- What advice would you give Simon to avoid repeating his mistakes in the future?
- What advice would you give to Tonya to avoid repeating her mistakes in the future?

d. Complete the table to describe the error(s) each of the other two students made.

Student's Name	Simon	Tonya
Error(s) Made When Evaluating	Substituted 7.5 instead of -7.5 for b Found the $\sqrt{49}$ instead of the $\sqrt{64}$ Subtracted 16.81 from 7 instead of subtracting 16.81 from 8 Found the answer -9.81 instead of the answer -8.81	Added 16.81 instead of subtracting 16.81 Found the answer 24.81 instead of the answer -8.81



12.7.4 Guided Practice: Evaluating Expressions With Radicals

1. Find the value of the expression. Show your work.

$$\begin{aligned} & \frac{18 - 2 \cdot \sqrt{121}}{2^3} \\ &= \frac{18 - 2 \cdot 11}{2^3} \\ &= \frac{18 - 22}{2^3} \\ &= \frac{-4}{2^3} \\ &= -\frac{4}{8} \\ &= -\frac{1}{2} \end{aligned}$$

2. Find the value of the expression $\frac{6.2 + \sqrt[3]{z}}{12}$, where $z = -27$.

$$\begin{aligned} &= \frac{6.2 + \sqrt[3]{-27}}{12} \\ &= \frac{6.2 - 3}{12} \\ &= \frac{3.2}{12} \\ &= \frac{4}{15} \end{aligned}$$

12.7.4 GUIDED PRACTICE: EVALUATING EXPRESSIONS WITH RADICALS

DIFFERENTIATE INSTRUCTION

Prompt students to make the connection between this question and the Pythagorean theorem, which we have been working on in the last few lessons.

TEACHER MOVES

Assess the needs of your students while monitoring them working through question 2 to determine if any students need a refresher on cube roots.

12.7.5 YOUR TURN!

TEACHER MOVES

If time is an issue, then consider assigning 12.7.5 Your Turn! for homework.



12.7.5 Your Turn!

Find the value of each expression. Show your work.

$$\begin{aligned}
 1. \quad & \frac{5 \cdot 3^4}{-\sqrt{225}} \\
 &= \frac{5 \cdot 81}{-\sqrt{225}} \\
 &= \frac{405}{-\sqrt{225}} \\
 &= \frac{405}{-15} \\
 &= -27
 \end{aligned}$$

$$\begin{aligned}
 2. \quad & \sqrt[3]{100 - 6^2} \\
 &= \sqrt[3]{100 - 36} \\
 &= \sqrt[3]{64} \\
 &= 4
 \end{aligned}$$

$$\begin{aligned}
 3. \quad & \left| 4^{-1} + \sqrt[3]{\frac{1}{8}} \right| \\
 &= \left| \frac{1}{4} + \sqrt[3]{\frac{1}{8}} \right| \\
 &= \left| \frac{1}{4} + \frac{1}{2} \right| \\
 &= \left| \frac{1}{4} + \frac{2}{4} \right| \\
 &= \left| \frac{3}{4} \right| \\
 &= \frac{3}{4}
 \end{aligned}$$

ENRICHMENT

Question 4 involves substituting a negative value for the variable and subtracting. Prior to completing question 4, pose questions to address this, such as:

- After substituting, will the value of w remain negative?
- After squaring, will the value of w remain negative?

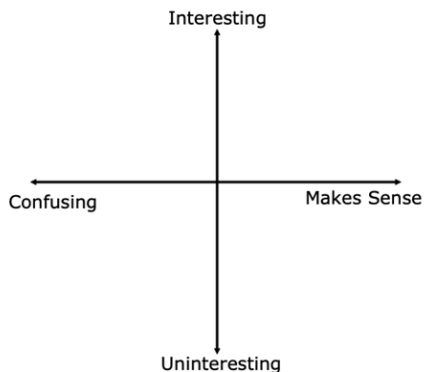
$$\begin{aligned}
 4. \quad & \frac{\sqrt{81 - w^2}}{9} + 25 \text{ where } w = -4 \\
 &= \frac{\sqrt{81 - (-4)^2}}{9} + 25 \\
 &= \frac{9 - (-4)^2}{9} + 25 \\
 &= \frac{9 - 16}{9} + 25 \\
 &= \frac{-7}{9} + 25 \\
 &= -\frac{7}{9} + \frac{25}{1} \\
 &= -\frac{7}{9} + \frac{225}{9} \\
 &= \frac{218}{9} \\
 &= 24\frac{2}{9}
 \end{aligned}$$

**12.7.6 Wrap-Up: Reflection**

1. Plot a point on the grid to indicate your level of interest and how you feel about today's learning target.

Answers will vary.

- ☐ I can solve problems involving the order of operations with rational numbers, including exponents and radicals.



2. Write one question you still have about this learning target after today's lesson.

Answers will vary. Is there an acronym, that includes radicals and absolute value symbols, to help me remember the correct order of operations?

12.7.6 WRAP-UP: REFLECTION**TEACHER MOVES**

Consider pairing students to discuss their reflections to see if peers can answer any of their questions.